Huawei Access Node HEB-ACC-75 Optical Transport Failure SOP (ALARM_4188)

Node: HEB-ACC-75 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4188

Description

This Standard Operating Procedure (SOP) addresses the root cause of the Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss) in the Huawei Access Node HEB-ACC-75, which is a critical failure domain. The failure occurs due to the degradation of the SFP/XFP transceivers or fiber cuts that disrupt the optical signal transmission between the node and the neighboring nodes. The Huawei Access Node HEB-ACC-75 is an Access Node that uses the Huawei Optical Transport Network (OTN) technology for transporting the optical signals. The OTN protocol stack is based on the ITU-T G.709 standard, which defines the Optical Transport Network (OTN) hierarchy and the mapping of client signals onto the OTN transport network. The SFP/XFP transceivers are used to transmit the optical signals between the nodes, and their degradation or failure can cause the signal loss, leading to the failure of the Optical Transport Network. The failure of the Optical Transport Network can cause the loss of critical services such as voice, video, and data transmission. Therefore, it is essential to identify and resolve the root cause of the Optical Transport Failure in a timely and efficient manner.

Pre-requisite Checks

- Verify that the Hebbal Edge site is accessible and safe for remote hands to perform the required actions.

- Check the redundant link status to ensure that the network is not affected by the failure of the primary link.

- Verify that the Huawei Access Node HEB-ACC-75 is properly powered and configured.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display device transceiver, Expected Output: Transceiver information, including the SFP/XFP type, status, and temperature, Description: This command displays the transceiver information, which can help identify the root cause of the Optical Transport Failure. The command output includes the SFP/XFP type, status, and temperature, which can indicate whether the transceiver is degraded or failed.
```

```
Command: display fiber optics, Expected Output: Fiber optics
information, including the fiber type, status, and length,
Description: This command displays the fiber optics information, which
can help identify the root cause of the Optical Transport Failure. The
command output includes the fiber type, status, and length, which can
indicate whether the fiber is cut or damaged.

Command: display optical power, Expected Output: Optical power
information, including the Rx and Tx power levels, Description: This
command displays the optical power information, which can help
identify the root cause of the Optical Transport Failure. The command
output includes the Rx and Tx power levels, which can indicate whether
the power loss is due to the SFP/XFP degradation or fiber cuts.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceivers, Command: replace transceiver {SFP/XFP ID}, Description: This step involves replacing the degraded or failed SFP/XFP transceivers with new ones. The command replaces the transceiver with the specified ID.

Step 2: Step: Repair or replace the fiber, Command: repair fiber {fiber ID}, Description: This step involves repairing or replacing the cut or damaged fiber. The command repairs or replaces the fiber with the specified ID.

Step 3: Step: Adjust the Rx/Tx power levels, Command: adjust optical power {Rx/Tx power level}, Description: This step involves adjusting the Rx/Tx power levels to compensate for the power loss. The command adjusts the power levels to the specified value.

Post-Check Verification

```
Command: display device transceiver, Expected Output: Transceiver
information, including the SFP/XFP type, status, and temperature,
Description: This command verifies that the transceiver is properly
configured and functioning correctly.

Command: display fiber optics, Expected Output: Fiber optics
information, including the fiber type, status, and length,
Description: This command verifies that the fiber is properly
configured and functioning correctly.

Command: display optical power, Expected Output: Optical power
information, including the Rx and Tx power levels, Description: This
command verifies that the power levels are within the specified range.
```

Roll-back Procedures

- Step: Rollback the transceiver replacement, Command: rollback transceiver {SFP/XFP ID}, Description: This step involves rolling back the transceiver replacement to the previous configuration. The command rolls back the transceiver to the specified ID.

- Step: Rollback the fiber repair or replacement, Command: rollback fiber {fiber ID}, Description: This step involves rolling back the fiber repair or replacement to the previous configuration. The command rolls back the fiber to the specified ID.

- Step: Rollback the power level adjustment, Command: rollback optical power {Rx/Tx power level}, Description: This step involves rolling back the power level adjustment to the previous configuration. The command rolls back the power level to the specified value.